AJAY VENKAT A

Data Scientist | Machine Learning Engineer

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PROFESSIONAL SUMMARY

AI & Data Science student with hands-on experience in Machine Learning, Deep Learning, Computer Vision and NLP. Delivered end-to-end AI applications such as real-time object detection and RAG systems, with 15+ projects on GitHub. Seeking a Data Scientist / ML role to build and deploy data-driven solutions.

INTERNSHIPS & EXPERIENCE

Quenoxa Technology - *Machine Learning Intern* Jun 2026 - Jul 2026

- Worked on machine learning tasks including data preparation, feature engineering and model evaluation

Transvaal Global Tech Pvt Ltd - *Technical Business Analyst Intern (AI, ML & Application Development)* Dec 2025 - Jan 2026

- Developed ML models in Python and Scikit-learn to support business decision-making
- Translated business requirements into AI solutions and application features

Techknots Pvt. Ltd. - *Power BI & Data Analytics Intern* Jun 2024

- Built Power BI dashboards with DAX and data modeling to visualize model and business metrics

FREELANCE & REAL-TIME PROJECTS

Public Grievance Portal for an MLA Office | *Full-Stack Web Application* github.com/ajay-venkat/tvk-portal

- Built a web portal where the public submits grievances directly to their MLA, with complaint tracking for the constituency office
- Delivered as a freelance project for a political party

Ration Shop Booking System (PDS) | *Web Application* Demo completed

- Built a booking flow so the public can reserve ration shop slots and collection in advance, reducing queues
- Demo completed; basis of my published IEEE research paper

Multi-Branch Bakery Ordering System | *Web / App Ordering Platform* 25+ branches

- Developing an ordering system for a bakery brand with 25+ branches under one platform

PROJECTS

VisionAid - Real-Time Navigation Assistant | *Python, YOLOv8, MiDaS, TTS, Cloud* github.com/ajay-venkat/visionaid

- Built a real-time navigation assistant using YOLOv8 for object detection and MiDaS for depth estimation
- Deployed as a cloud web application with text-to-speech audio guidance

Atlas RAG - Document Verifier | *Python, RAG, LLMs, Vector Search* github.com/ajay-venkat

- Built a retrieval-augmented generation (RAG) system that verifies documents against source content

Road Pothole Detection | *Python, YOLOv8, GPS / Geospatial Mapping* github.com/ajay-venkat

- Detected road potholes with YOLOv8 and mapped them with GPS coordinates for road-maintenance reporting

Market Basket Analysis | *Python, Pandas, Apriori, MongoDB, Full-Stack* github.com/ajay-venkat/market-basket-analysis

- Mined product association rules from retail transactions using the Apriori algorithm and Pandas

TECHNICAL SKILLS

Languages: Python
ML & DL: Scikit-learn, Deep Learning, Computer Vision, YOLOv8, MiDaS, NLP, RAG, Feature Engineering
Data & Libraries: Pandas, NumPy, Apriori, MongoDB, Power BI, DAX
Tools: Streamlit, GitHub, n8n, Make, Lindy, Cloud Deployment

ACHIEVEMENTS

- 1st Prize - National-level Mobile App Development Competition, India
- Participated in 5 hackathons: 2 First Prizes and 1 Second Prize
- Published research paper on a ration distribution flow system - IEEE, Scopus-indexed

CERTIFICATIONS

Agentic AI - Anthropic | Generative AI for Data Science - Microsoft | Google Cloud Cybersecurity - Google | AI & Python Development - Udemy

EDUCATION

B.Tech, Artificial Intelligence & Data Science 2023 - 2027

Karpaga Vinayaga College of Engineering and Technology (Affiliated to Anna University), Tamil Nadu CGPA: 8.05 / 10